

AI Usage Declaration

Project: Jamal Cart— AI-Powered E-Commerce Chatbot

Group Members:	Shahana Mansoor Jamal ----3 rd year BSCS---B23110006152
Declaration Date:	27th july, 2026

1. AI Tools Used

Claude (Anthropic), an AI coding/writing assistant, accessed via its web chat interface, was used substantially throughout the development of this project.

2. How It Was Used

Claude was used interactively across an extended,

- Generate some initial layered backend architecture (FastAPI routers, Pydantic schemas, service layer, database client) from a written specification we provided and some manual typing
- Generate the TensorFlow/Keras intent-classification training script and prediction module.
- Generate the chatbot's intent-routing logic (chatbot/actions.py), conversation-state management, and regex-based entity extraction.
- Generate the web frontend (HTML/CSS/JavaScript) and a Flutter mobile frontend against the same backend API.
- Diagnose and fix bugs we discovered through our own live manual testing (e.g. an address-extraction bug and a broken multi-turn add-to-cart flow, both identified by us from real chat transcripts and reported to Claude for a fix).
- Reconcile the application code against our actual Supabase database schema and seed data, which we supplied.
- Expand the chatbot's intent coverage (additional customer-support intents, persona, coupon/refund logic, email escalation) based on functional requirements we specified.
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All AI-generated code was reviewed, tested by running it against my real Supabase project, and iterated on based on our feedback before being included in the final submission.

3. Sections Written or Generated with AI Assistance

- Some Python source code and debugging: routers/, services/, schemas/, chatbot/, database/supabase.py, api.py
 - Web frontend: templates/index.html, static/css/style.css, static/js/app.js
 - Flutter frontend: all files under flutter_frontend/lib/(not tested yet)
 - chatbot/intents.json (intent categories and example patterns- different use cases)
 - This project report (Sections 1–15), the README.md, DATASET.md, and MODEL_CARD.md
- [Add/remove items here to accurately reflect your actual submission.]

4. Parts Implemented / Decided by the Group

This project was independently designed, developed, configured, tested, and validated by Shahana Mansoor Jamal. The following work was completed manually by the developer throughout the project lifecycle:

Project Planning

- Defined the overall project objectives and functional requirements.
- Designed the software architecture using a layered service-oriented approach.
- Planned chatbot capabilities, business rules, and customer interaction flow.
- Designed REST API endpoints and application workflow.

Database Design

- Designed the relational database schema.

- TensorFlow model development assistance
- FastAPI implementation assistance
- frontend integration guidance
- testing recommendations
- software engineering best practices

All AI-generated content was critically reviewed, verified, modified where appropriate, integrated into the project by the developer, and tested before being accepted.

The developer understands the complete architecture, implementation, algorithms, business logic, database design, and deployment process of this project and is able to explain every major component.

To the best of the developer's knowledge, this declaration accurately represents the extent of AI assistance used during the development of the project.

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